

INSTITUTE OF ACTUARIES OF INDIA

Subject SA3 - General Insurance

November 2024 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i) The aim of a financial plan is to have:

- One or more goals. Eg- increase the premium income by 10% per annum, write premiums of INR x of new products, double the market share in 5 years, increase the market share to y%.
- A strategy to be followed to achieve those goals
- A set of targets that will enable the success or failure of the strategy to be measured, eg- expected loss ratios (gross and net of reinsurance), new business volumes and renewal retention rates, capital implications, overall profitability of company moving forward
- Financial projections under multiple scenarios – optimistic, most likely, pessimistic
- Possible risks in achieving strategy and guard rails to be put in place

Gather and Analyze Data:

- Collect all the relevant financial data for Company Guardian, including historical financial statements (balance sheet, income statement, and cash flow statement), budgets, and any other financial projections or forecasts.
- Analyze the data to understand the company's current financial position, performance trends, and any significant financial events or changes that have occurred.
- Past data may not be useful for some of the new age evolving products and inputs from Underwriters, reinsurers and brokers may be required

Understand the Business and its Objectives:

- Familiarize yourself with the company's business model, industry, competitive landscape, and strategic goals.
- Engage with the management team to clearly define the company's financial objectives, such as revenue growth, profitability targets, cash flow management, or capital investment plans.

Develop Financial Projections:

- Create a financial model that incorporates the company's historical data, industry trends, and the defined financial objectives.
- Based on Risk Appetite of the insurer, develop projections for the company's future financial performance, including revenue, expenses, cash flows, and key financial ratios.
- Consider manpower requirements / new skillsets required/ training requirements
- Incorporate sensitivity analysis to account for potential changes in market conditions, operational factors, or other uncertainties that could impact the financial projections. This will help to understand capital requirements under pessimistic scenarios.
- Reinsurance arrangements can be planned based on regulatory / risk-based capital regime to help manage capital requirements.
- Cost-benefit analysis of multiple strategies can be done at evaluation stage to understand the optimum strategy to meet business objectives.
- Insurer may like to keep Plan B ready.

Assess Financing Requirements:

- Evaluate the company's current and future financing needs, such as working capital requirements, capital expenditures, debt repayment, or potential for new investments.
- Analyze the available financing options, including debt, equity, or a combination of both, and assess the costs and benefits of each option. Reinsurance arrangements may be tweaked in initial years to manage capital requirements.

Assess the external environment:

- Soft/hard market – Direct side
- Availability of reinsurance - additional capacity and at what cost
- Economic considerations – expected future demand, expected capital cost.
- Law / regulatory – expected changes in legislation / regulations.
- Climate – shift in weather patterns
- Regulatory changes or constraints
- Dependence on specialist brokers and relationship with them

Communicate and Obtain Feedback:

- Present the financial plan to the company's management team and other key stakeholders, such as the board of directors or lenders.
- Solicit feedback and incorporate any necessary adjustments or revisions to the financial plan based on the stakeholders' input and any changing circumstances.

Monitor and Update the Financial Plan:

- Establish a regular review process to monitor the company's actual financial performance against the projections in the financial plan.
- Update the financial plan periodically to reflect any changes in the company's objectives, market conditions, or other relevant factors that may impact the financial projections.

[Max 8]

ii) First party Covers financial loss from a cyber event to the insured party against:

- Damage to data and software- cost involved in recovering lost or corrupted data caused by malware attacks .
- Business interruption- financial protection for businesses in the event of a disruption to their normal operations. The main purpose of business interruption insurance is to help a business recover from the loss of income and increased expenses resulting from a covered event.
- Extortion - financial losses incurred by means of ransom or compensation paid to resolve cyber extortion.
- Reputational damage- legal costs, cost of removing objectionable content posted by cyber-bullies and psychological consultation costs.
- Theft of electronic funds- financial losses incurred in bank account, credit/debit cards, digital wallets arising from online frauds like unauthorized access, phishing, spoofing.

Third party cover covers financial consequence of any liability actions brought against the insured due to a cyber event. This includes:

- Security and privacy breaches
- Expenses to notify customers about a breach.
- Investigation
- Defence costs
- Damages arising from defamation, breach of privacy and intellectual property infringement.
- Loss of third-party data.

[Max 6]

iii) Coverage scope:

- Personal cyber policies typically cover individual consumer-level risks, such as identity theft, online fraud, cyber bullying, and the restoration of personal data.
- Commercial cyber policies are designed to protect businesses and organizations from a broader range of cyber-related risks, including data breaches, network interruptions, business interruption, cyber extortion, and liability claims.

Limits and deductibles:

- Personal cyber policies generally have lower coverage limits.
- Commercial cyber policies often have higher coverage limits with higher deductibles to match the scale and complexity of the business.

Underwriting and pricing:

- Personal cyber policies are usually underwritten based on individual risk factors, such as credit history and online activity. Typically, individual risks will be well within agreed terms and conditions of reinsurance treaty.
- Commercial cyber policies are underwritten based on the business's industry, size, revenue, data assets, and overall cybersecurity posture. May need special approval from reinsurer in some cases.

Insured entities:

- Personal cyber policies cover the individual policyholder and their immediate family members.
- Commercial cyber policies cover the business entity, its employees, and potentially other related third parties, such as vendors or contractors.

Risk management services:

- Commercial cyber policies often include additional risk management services, such as security assessments, employee training, and incident response planning, to help businesses proactively manage their cyber risks.
- Personal cyber policies may provide limited or no risk management services.

[Max 10]

iv) Regulatory and Compliance:

- Approval for the product from the PMC / regulator
- All areas of regulatory compliance for such a product – reinsurance arrangements, proposal forms, Customer Information Sheet, IT system etc.

Business side:

- Feasibility of business plan projection for such a product
- Projection of profitability and capital over the next 5 years
- Integration of distribution channels – largely specialist brokers
- Changes to agent remuneration, if any
- Changes in Key performance indicators, risk controls
- Review of business practices if some of them are not aligned with company's current practices.
- Underwriting expertise required.

Operation side and Systems and IT infrastructure:

- Review of system integration
- Development of systems for all key processes like UW, Reinsurance, Claims, Policy issuance and renewal, Customer service etc.

Actuarial side

- Review of current ALM position and expected gaps and opportunity in terms of diversification benefits and capital requirements.
- Calculation of portfolio risk adjusted return.
- Calculation of liability provisions
- Assessment of solvency position
- Pricing for such a product – expected to be reinsurance driven product.

Challenges in actuarial estimation- pricing, reserving, capital modelling:

- Traditional reserving method assumption - past data a good indicator of the future. Limited past claims data due to the low frequency and high severity nature of cyber insurance
- Fast-changing cyber risk landscape and technological advances – historical data less reliable for future predictions
- Long tail of the cyber claims due to the uncertainty in the settlement, third-party litigation.
- Long tail of claims arising from regulatory fines due to data breach violations, inadequate security measures, consumer protection violations, negligence in risk assessment, non-compliance with regulations
- Accumulation of risk:
 - Systemic nature of cyber-attacks
 - Interdependency structures between risks
 - Interconnected loss scenarios
- Complex dependency structures

Capital Modelling

- Risk Based Capital requirement for the new product.

Reinsurance

- Reinsurance arrangement for such product – minimum rate required by treaty, other treaty terms and conditions, overall capacity available, need for Risk XL/aggregate XL protection.
- Technical Support available / required for underwriting and claim settlement etc.
- Commission arrangements – will be an important parameter in case of lower retention.
- Rating and domicile of reinsurer

Finance

- Solvency projection and impact of selling the new product on topline and bottom line.
- How to consolidate the results with another line or report its results separately
- Impact of the new product on Profit and Loss statement and Balance Sheet
- Opportunities for cost rationalization
- Any future requirement of capital and sources available

Investments

- Any changes required in investment framework along with liquidity requirements.

Sales and Marketing

- Marketing strategies for the new product-whether it will go with existing channels or new channels.
- Engagement with Brokers

Risk

- Risk appetite – maximum risk that can be retained.
- Updating risk register with new risks and controls put in place.
- Review of overall risk governance and risk policies with respect to new product launches

[Max 10]

- v) Drone insurance is a specialized type of coverage that protects drone operators from liability and physical damage risks associated with operating drones.

It typically covers third-party liability, property damage, personal injury, and potentially loss or damage to the drone itself.

Drones are widely used for, example:

- Military and Defence
- Disaster response and management
- Aerial photography and videography
- Inspection of infrastructure (e.g., power lines, pipelines, construction sites)
- Precision agriculture (e.g., crop monitoring, spraying)
- Delivery of goods and packages
- Scientific Research and Exploration
- Drone racing and competitions.

Key coverages

- Liability coverage: This protects the drone operator from legal liability if the drone causes bodily injury or property damage to a third party.
- Hull/physical damage coverage: This covers the cost of repairing or replacing the drone in the event of physical damage, such as crash or theft.
- Payload coverage: This provides protection for any specialized equipment or cargo being carried by the drone.
- Privacy and data breach coverage: This covers legal liability if the drone's camera or other sensors are used to invade someone's privacy or if data collected by the drone is breached.

- Loss of use coverage: This provides compensation for the loss of use of the drone due to a covered event.

Regulatory Considerations:

- As per Indian drone rules, all drones except nano-category drones are required to compulsorily obtain third-party liability cover before commencing any kind of commercial operations. Failure to comply may result in legal action against the drone operator/owner.
- Ensure the drone insurance policy meets any mandatory coverage requirements.

Choosing a Drone Insurance Provider:

- Look for insurance companies that specialize in drone coverage or have experience in the drone industry.
- Compare coverage options, limits, deductibles, and pricing from multiple providers.
- Consider the reputation, financial stability, and customer service of the insurance company.
- Maintain proper documentation, including pilot certifications and drone registration, to support insurance claims.
- Review the policy terms and exclusions carefully to understand the coverage and any limitations.
- Consider higher coverage limits or additional policies if the drone is used for high-risk or commercial applications.

[Max 8]

[42 Marks]

Solution 2:

- i) Claim settlement ratio is the number of claims settled during the year from the total number of claims available for processing.

By insurer

Factors affecting claim settlement ratio include:

- Insurer with higher proportion of retail business can have higher claim settlement ratio – e.g. retail Health and Motor OD claims are settled quickly.
- Health business with OPD coverage will have higher settlement ratio.
- Mix of business Direct, Coinsurance - leader or follower – the way claims are booked can impact the settlement ratio. E.g. Leader can book individual claims of Group Health Policy while follower might be booking it as bulk entry with single count effectively impacting the settlement.
- efficient claims processing, customer-centric approach, and better financial resources.
- Larger, more established insurers often have higher claim settlement ratios as they have the infrastructure and experience to handle claims more effectively.
- Newer or smaller insurers may have lower claim settlement ratios as they are still building their claims management capabilities.
- Health - Cashless v/s non-cashless mode, higher proportion of cashless claims can result in higher settlement ratio.
- New-age or old established insurer – old player may have high proportion of litigation claims which may take long to settle.

- Claim settlement practices – Claim investigation criteria, documentation requirements, other claim servicing TATs and clarity of communication can have an impact of claim settlement ratio.
- Insurers In different economies, regulatory environment have different settlement ratios.
- Manpower / strength of claims department - Claim handling capacity, Claim handling expertise.
- Number of claims based on location, underwriting – impact of possible fraud element.
- Geographic locations- catastrophe, poor risks, - litigation experience may be linked to location and can lead to lower/ faster claim settlement.
- Distribution Channel – e.g. agents might be assisting in documentation while direct channel.
- Mix of business -e.g. mix of liability business. Higher proportion of liability business or Motor Third Party may lead to lower claims settlement ratio.
- business with poor claims experience or good experience – any cashflow problem can contribute towards slower claims settlement ratio.
- Terms and conditions related differences if company wordings differ from generally accepted industry wordings leading to higher rejections
- How on account payment is treated
- Lack of documentation from insured leading to delay in settlement
- Aggressiveness of company to pursue conciliatory settlement and settlement on non-standard basis.
- type of claims- large or attritional – typically large claims take more time to settle.
- catastrophe – typically claims will be settled with high focus and can result in higher settlement ratio.
- number of claims available for processing – any backlog from prior years
- rejection of claims

Different lines of business by insurers:

- Product type- needing loss adjustors, threshold for appointing surveyor
- Health insurance policies typically have higher claim settlement ratios compared to general insurance lines like motor, property, or liability.
- Motor TP insurance claims may have a lower settlement ratio due to the complexities involved in assessing and approving claims.
- Property insurance claims can also have a lower settlement ratio, as they may require more extensive investigation and documentation.
- Liability insurance claims, such as professional indemnity or directors and officers liability, may have a lower settlement ratio due to the subjective nature of establishing liability.
- Marine- delays in general average claims
- Usage of AI/ ML – e.g. Motor OD claims might be settled quickly with the help of technology with minimal human intervention
- Claim handling expertise between different lines varies
- Different coverages, policy conditions. Exclusions – e.g. Business interruption claims will take longer to settle or claims under Health Top- up policies can take longer to settle

- The claims settlement ratio may also be affected by the liability component of the segment written. For liability insurance, claims payment takes place in the future due to which immediate short-term settlement of claims is low
- Within a specific line of business, the claim settlement ratio can vary based on the type of claim.
 - For instance, in motor insurance, claims related to third-party liability may have a lower settlement ratio compared to own-damage claims.
 - In health insurance, claims for routine or minor treatments may have a higher settlement ratio than claims for complex or chronic conditions requiring longer hospital stay.

[Max 7]

ii) Factors:

- A significant increase in the number or cost of claims, due to catastrophic events or trends, can strain the company's financial resources.
- Setting premiums too low or inadequate pricing structure (or very different from competitors) can lead to insufficient funds to cover future claims, ultimately resulting in financial instability.
- Rapid growth with new products, new geographies or new distribution channels can result in anti-selection
- Refusal by reinsurer to pay their share (e.g. treaty not signed) or disagreement over claim payment
- Large coinsurance / reinsurance balances in receivables, Govt. dues
- Inadequate reinsurance protection (in case of large losses and events) can strain financial results
- Lack of Governance / failure of control functions – internal fraud, abuse etc.
- Wrong selling practices – required to refund money
- Large contingent liabilities
- Inadequate risk assessment and underwriting processes or controls can lead to a concentration of high-risk policies / business, increasing the likelihood of large losses.
- Failing to set aside enough reserves for future claims can result in cash flow issues when claims are filed.
- Inability to raise capital in timely manner
- Poor investment decisions or market downturns can erode the assets held by the insurer, impacting its ability to pay claims. E.g.- large proportion of assets in volatile asset class like equity
- Failing to adhere to regulatory requirements can lead to penalties, increased scrutiny, restriction on new business and potential license revocation
- Data breach
- High administrative costs in initial years with no economies of scale benefit kicking in
- inefficient claims processing or lack of claims investigation, and poor management practices can reduce profitability and increase the risk of insolvency.
- Broader economic challenges can reduce demand for insurance products and increase default rates on premiums.

Consequences:

1. Regulatory Consequences

- Regulatory intervention

- Regulator monitoring business plans, restrictions on writing new business
- Compliance Violations: If solvency is found to be overstated, the company may face legal repercussions, including fines and increased regulatory scrutiny.
- Revocation of Licenses: In severe cases, the company could lose its license to operate, which would halt all business activities.

2. Misallocation of Resources

- Overconfidence in Financial Position: Management may make overly ambitious investments or expansions based on inflated solvency figures, risking financial stability.
- Underestimating Risks: An overstated solvency can lead to inadequate risk assessment, resulting in the acceptance of high-risk policies that could lead to significant losses.

3. Market Perception and Trust Issues

- Loss of Credibility: Stakeholders, including policyholders and investors, may lose trust if the company's true financial health is revealed.
- Stock Price Volatility: If the market discovers the overstated solvency, it may lead to a sharp decline in stock prices and market capitalization.
- Rating downgrade of the Company

4. Operational Inefficiencies

- Inappropriate Risk Management: The company may not allocate sufficient resources to risk management, leading to vulnerabilities that could result in significant losses.
- Ineffective Business Strategies: Overconfidence in solvency can result in poor strategic decisions that do not align with actual financial capabilities.

5. Impact on Policyholders

- Claims Handling: If solvency is later revealed to be lower than stated, the company may struggle to fulfil claims, leading to dissatisfaction and potential legal issues.
- Premium Adjustments: The company might need to increase premiums abruptly to stabilize finances, affecting customer retention.

6. Long-term Viability

- Sustainability Risks: Overstated solvency can mask underlying financial issues, jeopardizing the long-term viability of the company.
- Impact on Growth: Future growth could be hindered if the company has to dedicate resources to rectify solvency issues instead of investing in new opportunities.

[Max 10]

[17 Marks]

Solution 3:

- i) A thorough analysis of reserves is essential to ensure financial stability and regulatory compliance. Factors which can be considered .

1. Reserve Adequacy

- Actuarial Review: Conduct a thorough actuarial review to assess the adequacy of reserves for both entities, considering historical claims data and future projections.

- Target companies tend to dress up the financials prior to acquisition by reserving aggressively or releasing reserves from past book. Eg- any trend of aggressive claims closure and re-opening of claims later
- Loss Development Factors:
 - Analyze historical claims data from Beta to identify trends in claims development.
 - Understanding how claims have evolved over time is crucial for accurately estimating reserves
 - Under staffing or degrowth in business volumes, change in business mix can cause change in claims paid pattern and claims reporting pattern
- 2. Product Mix and Risk Profile:
 - Review the types of insurance products offered by Beta. Different products carry varying levels of risk, which must be factored into reserve calculations.
 - Reasonableness checks on ULR assumption- in comparison with industry, in line with business written.
 - Features like Auto increase in SI or auto reinstatement options, new/ renewal mix, loss occurring / claims made policies, leader/follower/ RI inward business mix, any Non proportional business written can impact reserve estimation.
- 3. Actuarial Methodologies:
 - Evaluate whether Beta used similar actuarial methods for reserving. Differences in methodologies may require adjustments to align with Alpha's practices.
 - Going forward Combined claims triangle will require additional judgement as over time mix of business from two entities could have changed significantly – this can add more noise in development pattern.
 - Or reserving exercise can be performed separately and can be aggregated for some time before going for combined claims development to estimate the reserves until the practices between the two are aligned. The actuary can decide for how long to do it separately after the merger.
- 4. Integration Challenges
 - Cultural Differences: Assess how differing company cultures may affect claims management and reserve practices. Approach to claim settlement, default reserve strength, investigation criteria, reserving of litigation claims, treatment of orphan claims could vary between two insurers and for an insurer over time.
 - Systems Integration: Consider the challenges of integrating IT systems that track reserves and claims, which may affect data accuracy.
 - How coinsurance to be treated?
 - Ensure that the data from Beta aligns with Alpha's standards. Inconsistent data can lead to inaccurate reserve estimates.
 - Few of the data adjustments might not be available at claim level (e.g. reinsurance recoveries under Aggregate XL, Commutation arrangements etc.)
 - Different IT systems might be used across product categories by the combined entity.

5. Claims Experience

- Claims History: Compare claims history and trends for both companies to identify any discrepancies in claims handling or reporting.
- Emerging Claims: Evaluate any emerging claims and potential liabilities that may not be fully reflected in existing reserves.
- Claims assessors from Alpha will need to get familiarise with IT systems and product structure (terms and conditions, limits, deductibles etc.) of Beta

6. Regulatory Compliance

- Local Regulations: Ensure that reserves meet the regulatory requirements of the jurisdictions where the combined entity operates.
- Reporting Standards: Adhere to accounting standards (e.g., IFRS, GAAP) applicable to insurance reserves.
- Establish a clear governance framework for reserve management. Define roles and responsibilities to maintain transparency and accountability.

7. Market Conditions

- Economic Environment: Analyze the economic conditions that may impact claims frequency and severity, such as changes in regulations, market competition including new entrants, or economic downturns.
- Industry Trends: Stay informed about broader industry trends that could influence reserve requirements, such as shifts in underwriting practices or new types of coverage or any landmark judgement impacting reserve strength.

8. Reinsurance Considerations

- Reinsurance Treaties: Review existing reinsurance arrangements for both companies to understand how they impact reserve calculations.
- Counterparty Risk: Assess the financial health of reinsurers, as this can affect the reliability of recoverable.

9. Claims Management Practices

- Claims Handling Procedures: Evaluate the claims handling processes of both companies to identify best practices and areas for improvement.
- Fraud Detection: Consider the effectiveness of fraud detection mechanisms, which can impact reserve levels.
- Claims Processing Efficiency- Assess the efficiency of Beta's claims processing. Delays in claims settlements if not adjusted could result in potential under-reserving (e.g. understaffed claims team or issues in IT systems leading to more manual work).

10. Future Projections

- Growth Projections: Analyse expected growth in premiums and claims, as this will influence future reserve needs.
- Loss Forecasting: Develop forecasting models that incorporate both historical data and future expectations for loss trends.

11. Stakeholder Engagement:

- Prepare to communicate the results of your reserve analysis to key stakeholders, including the board, regulator, Peer reviewer and investors. Clear communication will help build trust in the financial health of the combined entity.

[Max 13]

ii)

- Lower the credibility of data, higher will be the risk adjustment percentage and vice-versa. To the extent that emerging experience reduces uncertainty about amount and timing of cash flows, risk adjustment will decrease. Example- for commercial lines such as marine, there is lesser data available to reliably estimate the trend.
- Risks with low frequency and high severity will result in higher risk adjustment than risks with high frequency and low severity. Example- motor own damage which has high frequency and low severity of claims is expected to have lower risk adjustment than engineering which has low frequency and high severity of claims.
- Risk adjustment increases with increase in the term of contract and tail of the business. Example- motor third party is expected to have higher adjustment due to longer tail compared to motor own damage.
- Risks with wider probability distribution will result in higher risk adjustment than risks with narrower distribution. E.g.- fire line of business has wider distribution as fire in a dwelling would result in a small claim but a fire in factory could result in a very large loss- this is expected to have higher risk adjustment compared to motor own damage which is expected to have stable probability distribution.
- The lesser that is known about the current estimate and trend for a line of business, higher the risk adjustment will be. Eg- for volatile lines such as liability lines, very less is known about current estimate and trend hence expected to have higher risk adjustment compared to health where there is greater knowledge about current estimate and its trend.
- Risk adjustment on ceded reinsurance will be based on the reinsurance structure
- Diversification impact of risk adjustment depending on the correlation between the lines of business can be considered.

[Max 5]

- iii) At initial recognition, if the contracts are expected to be loss-making/onerous based on experience of changes in circumstances, the loss component needs to be immediately recognised at initial recognition. If the contract is profitable then CSM is released with the provision of services over time.

The CSM under this contract of ABC is INR 1,00,00,000.

Assumptions

- The contract is measured under GMM approach as there is no CSM under PAA approach.
- The expected cashflows under this contract are assumed to be Premium, claims and expenses. The risk adjustment is included in the expected claims and expenses provided.
- The cashflows provided are discounted.

The CSM is released over time is released based on coverage units. Coverage units represent the quantity of insurance coverage provided by an insurance contract over expected duration.

For this line of business- ABC, we can consider the uniform earning over 5 years as the coverage unit release pattern hence the expected release pattern of CSM at inception for this contract is:

Year 1-INR 20,00,000

Year 2-INR 20,00,000

Year 3- INR 20,00,000

Year 4- INR 20,00,000

Year 5- INR 20,00,000

[Max 5]

iv)

- The cash flows for reinsurance need to be adjusted for credit risk/default risk from reinsurer.
- the contract boundaries for risk attaching treaties may be wider than the corresponding direct contract boundaries, for example- in case of fire contracts reinsured on risk-attaching basis
- Contracts on loss-occurring basis with term of less than or equal to one year can be run via PAA but risk-attaching treaties will require to be run via GMM/PAA eligibility test since the contractual term is more than one year
- For non-proportional contracts, the reinsurance contract will be recognised either at start of reinsurance contract or the date the entity recognises an onerous group of underlying contracts, whichever is earlier. In case of proportional reinsurance, the contract will be recognised either at start of coverage of RI contract or initial recognition of gross contract, whichever is later.
- A single reinsurance treaty might provide cover for multiple portfolios. Alternatively, multiple RI arrangements might cater to the same portfolio. A decision will need to be made on how to account for these
- For proportional reinsurance (eg- quota share), company may use same percentage of risk adjustment as gross cohort, however it may need to recalculate the risk adjustment for non-proportional reinsurance held
- For reinsurance, loss component is not booked in same manner as gross contracts. However, recovery component of gross loss component would be created.
- Level of aggregation may differ from Direct portfolios.
- Disclosures required for reinsurance contracts

[Max 5]

v) Functionality Requirements

- Discuss the core capabilities that the tool must possess, such as:
 - Pre-defined data model
 - Pre-defined data processing engines
 - Flexibility for customization
 - Data processing at scale
 - Inbuilt IFRS and subledger modules
 - Predefined IFRS cost of Acquisition
 - Predefined IFRS reports

Data Integration and Usability

- Seamless Data Integration:
- Emphasize the importance of the tool's ability to integrate with existing systems, such as the accounting software and ERP systems used by Insurer Alpha. Discuss the potential inefficiencies and errors if data needs to be manually transferred.
- User-Friendliness:-Highlight the need for an intuitive interface to configure inputs/outputs and the tool. A tool that is difficult to navigate could lead to user errors and decreased productivity. You propose conducting user testing to evaluate the tool's usability.
- It should have the ability to define rules
- It should have ability to configure workflows

Vendor health profile

- Financial health
- Market penetration
- Business partnerships/alliances
- Location- whether office in India?
- Support structure models (is it available 24*7) or only on business days
- Consulting hours provided

Skillset

- Market availability
- Trainings and certifications availability
- Activity on support groups/forums
- Documentation availability
- Usage of open-source codes

Commercial and contracting

- Total cost of ownership
- Pricing model (subscription or user/core-based licenses)
- Lock-in clauses
- Exit clauses
- Liabilities
- Legal protection

Deployment models

- Available both on-premises/cloud or only cloud
- Flexibility in compute/storage resources provisioning
- Speed to provision/deprovision resources
- Disaster Recovery
- High availability
- Severity level adherence
- Data preservation expectations

[Max 5]
[33 Marks]

Solution 4:

A thorough understanding of data flaws is essential for effective pricing in general insurance. It ensures that the pricing process is grounded in reliable information, leading to informed decisions, regulatory compliance, and ultimately, the long-term sustainability of the insurance company.

- Need to know the potential variability in the results produced from analysing the data. The product's premium can be compared with similar competitive products to help in quantifying the impact of data flaws.
- Different data problems will affect the results in different ways and to a varying extent. Understanding the nature of the problems will help in determining what allowance must be made.
- Helps in determining the cause of the data problems which can in turn help to decide what actions to take to prevent such problems arising again in the future. Helps in finding improvements to the data capture methods.
If key product of insurer, then incorrect pricing due to inaccurate data can have major impact on solvency. Eg- incorrect pricing can lead to higher sales driving up RSM and losses leading to dip in ASM thus making the product non- sustainable in the long run.
- Professional standards such as:
 - APS 21 – Data Quality standards and what should be done in case of data quality issues
 - APS 34- including Data quality checks to ensure sufficient and reliable data, data validation, data modifications, deficiencies in data.

Accurate Pricing

- Risk Assessment: Accurate data is essential for assessing risk accurately. Flawed data can lead to incorrect premium calculations, resulting in either underpricing (leading to losses) or overpricing (leading to loss of customers).
- Loss Projections: Errors in data can distort loss projections, affecting the insurer's ability to estimate future claims accurately.

Regulatory Compliance

- Adherence to Standards: Regulators often require accurate data for pricing models. Inaccuracies can lead to non-compliance.
- Transparency: Understanding data flaws and how they are addressed helps ensure transparency in the pricing process, which is essential for regulatory reporting and audits.

Informed Decision-Making

- Strategic Decisions: Insurers rely on data for strategic decisions regarding product development, market entry, and competitive positioning. Flawed data can lead to misguided strategies.
- Resource Allocation: Accurate data helps in effectively allocating resources, such as marketing and underwriting efforts, based on reliable insights.
- Data may not be available by rating factors and by add-on to correctly do the repricing. Relativities chosen may be based on inputs from underwriters or reinsurer but may not be suitable over time. This can lead to future losses due to incorrect pricing grid resulting in cross subsidies / anti- selection.
- Pricing loss ratios may be used for setting initial reserves- data flaws can lead to reserving errors

Customer Trust and Satisfaction

- Fairness: Customers expect fair pricing. Competitors using accurate data may be able to charge lower premiums and pass on better rates to customers hence data Inconsistencies can lead to dissatisfaction and damage the insurer's reputation.
- Claim Handling: Understanding data errors helps ensure that claims processing and payouts are based on accurate information, enhancing customer trust.

Model Integrity

- Statistical Validity: Pricing models rely on historical data to predict future outcomes. If not handled appropriately, data flaws can compromise the statistical validity of these models, leading to poor results.
- Sensitivity Analysis: Identifying data inconsistencies allows for better sensitivity analysis, helping to understand how changes in data affect pricing outcomes.

[Max 8 Marks]
